Final Project Instructions

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Final Projects

Each group will prepare and submit a single zip file including three folders of:

1. **A paper** written in CVPR format on overleaf.com. Needs to be submitted in LaTex format including all the necessary files (you can use overleaf.com as a group). You can download the CVPR template from: https://media.icml.cc/Conferences/CVPR2023/cvpr2023-author_kit-v1_1-1.zip

2. **A presentation** file (in .ppt or .pptx format) to be presented to the instructor (about 15-minutes long presentation will be given by all the group members. **The missing group members will receive zero credit/point automatically**). You will be asked questions during your presentation.

3. **A GitHub page** for the project (include the code, a nice description, the dataset or a link to dataset). If the links or files do not work on the GitHub page, students may not receive credit for the project. It is students' responsibility to make sure that the GitHub page contains clear definition and all the necessary files to be run.

Deadlines & Presentation Time

- Each group must submit all project materials, including **(1)** a conference paper, **(2)** a PowerPoint presentation, and **(3)** a link to the project's GitHub repository, on Canvas by **11:59 PM on May 13, 2026.**
  - Please ensure that the GitHub link for your project is included **on the first page of your presentation along with the full names of all the group members**.
  - All documents must be submitted **by the evening of May 13** to be eligible to present your project. No modifications to any files, including the presentation, will be permitted after submission on Canvas. Non-compliance with these instructions will result in a 30% penalty per day of delay. Additionally, if the GitHub link is non-functional or inaccessible, the same penalty will apply. It is the group's responsibility to ensure the GitHub link is working and accessible to everyone.
- Each group will deliver **a 15-minute long presentation** to the instructor, with time slots allocated on a first-come, first-served basis. **Please select a time slot on either May 14 or May 15**, depending on the instructor's availability. Presentations should be prepared to accommodate technical questions.
- To request a time slot, one representative from your group should send an email listing the preferred presentation times and dates (**at least three options in 15-minute intervals**, starting from 11:00 AM and ending at 8:00 PM, should be listed). If your first choice is available, it will be assigned; otherwise, the next available option will be provided. Be sure to include your group name, the full names of all group members, and the title of your final project in the email.
- If your group is ready to present **before May 14th**, please email me to arrange an earlier presentation time.

Submission process of all your final documents

- In a single .zip file include:
  - your paper files (PDF file AND all its Latex source files in a folder)
  - Your PPT file that you will present at the end (in .ppt or .pptx format)

and submit that .zip file **on Canvas** (one submission per group is needed).

with the subject: "ECE4990_FinalProjectFiles_Group**<NAME>**" where you replace **<NAME>** with your group **name**.

- Include all your group members' name in your submission too (at the beginning of your powerpoint slides and on the first page of your final report). You can link **google drive** if your files are too large (for example for some datasets and/or for some trained models) to include in your zip file.

Paper & Presentation Structure

- **Abstract**: ideally, you should have already written this part at the beginning of the semester or after your paper presentation.
- **Introduction**: You should finalize this section at the end, after finalizing your experiments.
- **Related Work**: You should have already done this part during your survey (paper) presentation. Just summarize those papers that you have presented during your paper presentation.
- **Proposed Approach (MOST IMPORTANT SECTION)**
  - **Architecture details (include its figures and relevant tables), Loss function (define and describe it) and input and output forms**
  - **The technical explanation of your main novelty (this cannot be hyper-parameter tuning and that is why you are asked to meet me /update me frequently).**
  - **Include your baseline architecture's details as well here.**
- **Results / Experiments (ANOTHER MOST IMPORTANT SECTION :)**
  - **Data: D**etails of the dataset that you used (resolution, # of training & test samples, input & output format, total # of classes, etc.).
  - **Metrics:** The definitions of the used metrics in your performance comparison. Define them as equations (make sure to explain the meaning of each used term in all the equations after giving that equation).
  - **Results:** Explain what tests/experiments you run to check the performance of your network. Compare your results to **at least** one baseline network that is the state of the art and on at least two different datasets. (Also check the syllabus).
- **Conclusion:** Conclude your work in a few paragraphs. Also in another paragraph, state the contribution of each group member (If you cannot agree on that, you can talk to me individually).
- **References:** Include all the relevant references. The more citations/references, the better.
- You can have as many pages as required (**Min. 9 pages in CVPR format without the references**). Make sure you do not leave a lot of spaces between texts, figures, equations ,etc. Also **make sure that you use your own figures, results and drawings** in your paper.
- The figures and plots should be your own drawings mainly. Do your best to refrain from using existing figures from other people's work. That has no contribution (value) to your own work.
- You can also have a look at your preferred conference or journal papers to get ideas about how to structure your own paper, how to present your results and how to describe your novelty.
- Note that (as usual), your presentation quality will also be considered, when grading.

What will be graded during your presentation

All the of the listed aspects will be considered when assessing your final projects and determining your final project grade:

- How well you followed the instructions,
- How well you presented the project,
- The degree of your completion of your final project, (whether you completely achieved all your goals or not),
- The depth of analysis and technical depth that you present,
- The detail and level of results you present.
- In your report and in your final presentation, an analysis on the effect of certain hyperparameters on the final performance is expected. (For example, how the performance change as you change the learning rate, how the performance change as you change the training batch size, etc.).

- You are expected to write your own final report and presentations.
- No help from third-party apps, websites or any form of AI based software is allowed.
- The technical depth of your understanding of your chosen field is also graded. For example:
  - if you are working on object detection algorithm: YOLO, then you are expected to know the architectural details (as much as possible) of the YOLO algorithm. You should be able to modify the architecture's certain aspects, such as adding or removing a layer, modifying its loss function, etc.